Table 24: Instruction examples for different tasks.

Instruction	Example
Lowercase	Please ensure that your response is in English, and in all lowercase letters.
Language	Please respond using only the German language, no other language is allowed.
JSON Format	Wrap the entire output in JSON format.
English Capitalize	Make sure your entire response is in English, and in all capital letters.
Sentence Frequency	Answer with exactly 3 sentences.
Word Inclusion	Please include the word cat in the response.

Few-shot Chain-of-Thought Examples

Question: Olivia has $23. She bought five bagels for $3 each. How much money does she have left? Answer: Olivia had 23 dollars.
She bought 5 bagels at 3 dollars each: 5 × 3 = 15.
She spent 15 dollars, so she has 23 - 15 = 8 dollars left.
The answer is 8.

Question: There were nine computers in the server room. Five more computers were installed each day, from monday to thursday. How many computers are now in the server room?
Answer: There were originally 9 computers.
5 computers were added each day for 4 days: 5 × 4 = 20.
So, total is 9 + 20 = 29.
The answer is 29.

Question: There are 15 trees in the grove. Grove workers will plant trees in the grove today. After they are done, there will be 21 trees. How many trees did the grove workers plant today?
Answer: There were originally 15 trees.
After planting, there were 21 trees.
So, 21 - 15 = 6 trees were planted.
The answer is 6.

Question: If there are 3 cars in the parking lot and 2 more cars arrive, how many cars are in the parking lot?
Answer: There were 3 cars.
2 more cars arrived: 3 + 2 = 5.
The answer is 5.

Question: Jason had 20 lollipops. He gave Denny some lollipops. Now Jason has 12 lollipops. How many lollipops did Jason give to Denny?
Answer: Jason had 20 lollipops.
After giving some away, he had 12.
So he gave away 20 - 12 = 8.
The answer is 8.

Question: Shawn has five toys. For Christmas, he got two toys each from his mom and dad. How many toys does he have now?
Answer: Shawn had 5 toys.
He got 2 from mom and 2 from dad: 2 + 2 = 4.
So he has 5 + 4 = 9 toys.
The answer is 9.

Question: Leah had 32 chocolates and her sister had 42. If they ate 35, how many pieces do they have left in total?
Answer: Leah had 32 and her sister had 42: 32 + 42 = 74.
They ate 35: 74 - 35 = 39.
The answer is 39.

Question: Michael had 58 golf balls. On tuesday, he lost 23 golf balls. On wednesday, he lost 2 more. How many golf balls did he have at the end of wednesday?
Answer: Michael started with 58.
Lost 23 on Tuesday: 58 - 23 = 35.
Lost 2 more on Wednesday: 35 - 2 = 33.
The answer is 33.

H Simple Evaluation/Verification with Visualization

Neuronpedia (Lin, 2023) is an interactive web-based platform that hosts labeled SAE feature dictionaries for open-weight models. Each SAE feature is tagged with human-interpretable descriptions (e.g., "list start", "closing brace", "polite language") based on automated annotation pipelines by gpt 4o. We utilize Neuronpedia to investigate and verify the semantics of key SAE latents identified during steering experiments. Specifically, we validate whether activation-based interventions correspond to meaningful internal concepts.

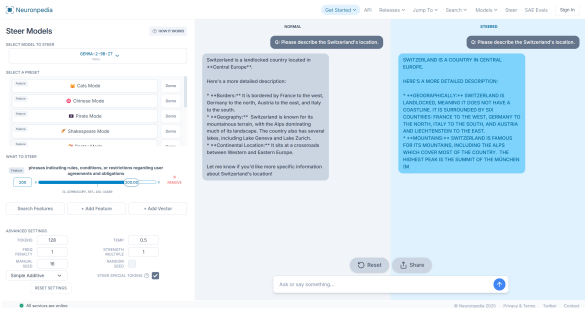

Figure 21: Example 1 by Neuronpedia: Steering models to generate uppercase answers by SAE steering vectors

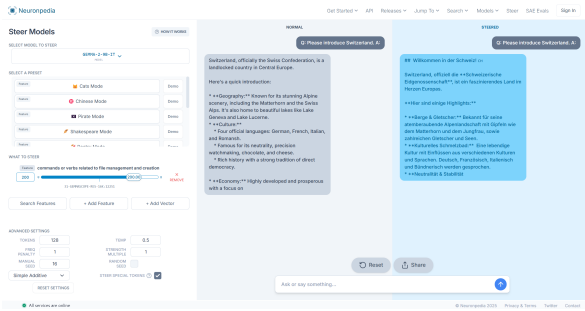

Figure 22: Example 2 by Neuronpedia: Steering models to generate German answers by SAE steering vectors

I Linear Probing with SAE Feature Gating

I.1 Methods

To quantitatively assess whether specific SAE features encode concepts related to arithmetic, we employ a linear probing methodology. The objective is to train a simple linear classifier to decode a specific arithmetic operation (addition, subtraction, multiplication, or division) from the internal representations of a language model. The performance of this probe serves as a measure of how linearly separable the target information is within the model's activations.

Our probing setup introduces a key modification to the standard approach by using an SAE feature vector as a gating mechanism. For a given input sequence representing an arithmetic expression, we first extract the sequence of hidden state activations, $\mathbf{H}$, from a target layer of the frozen language model. Instead of probing these activations directly, we modulate them via an element-wise Hadamard product ($\odot$) with a chosen SAE feature vector, $\mathbf{d}_{l,j}$:

$$\mathbf{H}_{\text{gated}} = \mathbf{H} \odot \mathbf{d}_{l,j} \tag{9}$$

This operation selectively amplifies or suppresses dimensions in the activation space that are aligned with the feature captured by the SAE vector.

The resulting sequence of gated activation vectors, $\mathbf{H}_{\text{gated}}$, is then aggregated into a single vector representation through mean-pooling across the token dimension. This final vector is fed into a linear classifier, which is trained to output logits for the four operation classes. A softmax function is applied to these logits to produce a probability distribution. The classification accuracy of this probe indicates the extent to which the specific SAE feature $\mathbf{d}_{l,j}$ encodes information relevant to the arithmetic operation being classified.

I.2 Results

Table 25: Probing with SAE 15153 and Gemma-2-2B

	precision	recall	f1-score
+	0.33	0.41	0.36
-	0.23	0.29	0.26
*	0.55	0.21	0.31
/	**0.72**	**0.82**	**0.76**
macro avg	0.46	0.43	0.42

Table 26: Probing with SAE 1642 and Gemma-2-2B

	precision	recall	f1-score
+	**0.44**	**0.56**	**0.49**
-	0.29	0.19	0.23
*	0.38	0.36	0.37
/	0.42	0.47	0.44
macro avg	0.38	0.40	0.38

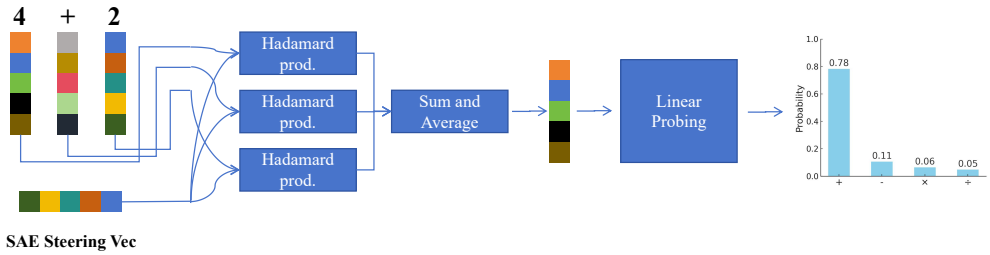

Figure 23: Linear Probing

Table 27: Probing with SAE 435 and Gemma-2-2B

	precision	recall	f1-score
+	0.77	**0.87**	**0.82**
-	0.55	0.43	0.48
*	0.50	0.52	0.51
/	**0.78**	0.83	0.81
macro avg	**0.65**	**0.66**	**0.65**

Table 28: Probing with MeanActDiff Steering Vector and Gemma-2-2B

	precision	recall	f1-score
+	0.30	0.11	0.16
-	0.43	0.89	0.58
*	0.72	0.14	0.23
/	**0.67**	**0.94**	**0.78**
macro avg	0.53	0.52	0.44

Table 29: Probing with SAE 7007 and Gemma-2-9B

	precision	recall	f1-score
+	0.41	0.76	0.53
-	0.11	0.05	0.07
*	0.78	0.33	0.47
/	**0.71**	**0.94**	**0.81**
macro avg	**0.50**	**0.52**	**0.47**

Table 30: Probing with SAE 9005 and Gemma-2-9B

	precision	recall	f1-score
+	0.34	0.75	0.47
-	0.42	0.15	0.22
*	0.46	0.04	0.07
/	**0.67**	**0.91**	**0.77**
macro avg	0.47	0.46	0.38

Table 31: Probing with SAE 6782 and Gemma-2-9B

	precision	recall	f1-score
+	0.45	0.75	0.56
-	0.46	0.25	0.33
*	0.36	0.17	0.23
/	**0.67**	**0.89**	**0.77**
macro avg	0.49	**0.52**	**0.47**

Table 32: Probing with MeanActDiff Steering Vector and Gemma-2-9B

	precision	recall	f1-score
+	0.45	0.75	0.56
-	0.24	0.15	0.18
*	0.72	0.22	0.34
/	**0.57**	**0.80**	**0.67**
macro avg	0.49	0.48	0.44

As shown in Tables 27 through 32, the linear